

1. 【简答题】

AC equivalent circuit of a single small signal resonant amplifier is shown as Fig.1. The resonant frequency $f_0 = 10\text{MHz}$, pass band $B_{0.7} = 500\text{ kHz}$, resonant voltage gain $A_{u0} = 100$. The tested Y parameters at operating point and operating frequency are as follows:

$$y_{ie} = (2 + j0.5)\text{ mS}, y_{re} \approx 0, y_{fe} = (20 - j5)\text{ mS}, y_{oe} = (20 + j40)\text{ }\mu\text{S}.$$

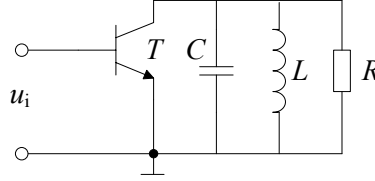


Fig. 1

答案:

$$\therefore A_{u0} = \frac{|y_{fe}|}{g_{\Sigma}} \Rightarrow g_{\Sigma} = \frac{|y_{fe}|}{A_{u0}} = \frac{\sqrt{20^2 + 5^2}\text{ mS}}{100} = 2.07 \times 10^{-4}\text{ S}$$

$$Q_L = \frac{f_0}{B_{0.7}} = \frac{10\text{MHz}}{500\text{KHz}} = 20$$

$$g_{\Sigma} = \frac{1}{Q_L \omega_0 L} \Rightarrow L = \frac{1}{Q_L \omega_0 g_{\Sigma}} = \frac{1}{20 \times 2\pi \times 10^7 \times 2.07 \times 10^{-4}} \approx 3.84\text{ }\mu\text{H}$$

$$C = \frac{1}{4\pi^2 f_0^2 L} = \frac{1}{4\pi^2 \times 10^{14} \times 3.84 \times 10^{-6}} \approx 66\text{ pF}$$

$$g_{\Sigma} = \frac{1}{Q_L \omega_0 L} + \frac{1}{R} + g_{oe} \Rightarrow \frac{1}{R} = 2.07 \times 10^{-4} - \frac{1}{60 \times 2\pi \times 10^7 \times 3.84 \times 10^{-6}} - 20 \times 10^{-6}$$

$$R \approx 8.48\text{ k}\Omega$$

2. 【简答题】

A single resonant amplifier with $|A_{u0}| = 20$, $B = 6\text{ kHz}$ at resonant frequency. If increasing the level of the same amplifier, please calculate the sum resonant voltage gain and that of pass band. If the sum of the pass band holding 6 kHz , please calculate the single level voltage gain A_{u1} .

答案:

(1)

$$A_{u\Sigma} = A_{u1} \times A_{u2} = 20 \times 20 = 400$$

$$B_{0.7\Sigma} = \sqrt{2^{\frac{1}{n}} - 1} \times B_{0.7} = \sqrt{\sqrt{2} - 1} \times 6\text{kHz} \approx 3.86\text{kHz}$$

(2)

$$B_{0.7\Sigma} = 6\text{kHz} = \sqrt{\sqrt{2} - 1} \times B_{0.7}$$

$$B_{0.7} \approx 9.33\text{kHz}$$

根据放大器增益带宽积为常数

$$400 \times 3.86 = A_{u\Sigma} \times 9.33$$

$$A_{u\Sigma} \approx 165.6$$

$$A_{u1} = A_{u2} = \sqrt{A_{u\Sigma}} = 12.87$$

3. 【简答题】

A resonant power amplifier operating at critical point state, if the equivalent load resistor R_c changing suddenly. If R_c increase double or reduce by half. How to change of output power P_o . Please analyze the reasons of changing.

答案:

(1) 其它条件不变时, 集电极等效负载电阻 R_c 加倍, 功放工作到过压状态, 输出功率会降低; 原因是输出基波电压缓慢增加, 而集电极电流顶部凹陷, 随着过压加深, 凹陷明显, 电流有效值迅速减小, 故输出功率减小。集电极效率先增加后减小。

(2) 其它条件不变时, 集电极等效负载电阻 R_c 减半, 功放工作到欠压状态, 输出功率也会降低; 原因是输出电压基波分量会迅速减小, 虽然集电极电流基波分量会缓慢增加, 但二者乘积减小, 故输出功率减小。集电极效率单调减小, 管耗增加, 严重时会烧毁功放管。

4. 【简答题】

A resonant power amplifier with operating at critical point state. Assuming $E_c = 24V$, current conduction angle (CCA) $\theta = 70^\circ$, $\alpha_0(70^\circ) = 0.253$, $\alpha_1(70^\circ) = 0.436$, collector current pulse maximum value $I_{c\max} = 2.2A$, collector voltage utilization factor $\xi = 0.9$. Please calculate output power P_o , voltage source providing power P_s , collector efficiency η_c and collector equivalent load resistor R_c .

答案:

$$U_{c1m} = \xi \times E_c = 0.9 \times 24 = 21.6V$$

$$I_{c1m} = \alpha_1(\theta) \times I_{c\max} = 0.436 \times 2.2 \approx 0.959A$$

$$P_o = \frac{1}{2} I_{c1m} U_{c1m} = \frac{1}{2} \times 21.6 \times 0.959 \approx 10.36W$$

$$I_{c0} = \alpha_0(\theta) \times I_{c\max} = 0.253 \times 2.2 = 0.5566A$$

$$P_s = E_c I_{c0} = 24 \times 0.5566 \approx 13.36W$$

$$\eta_c = \frac{P_o}{P_s} = \frac{10.36}{13.36} \approx 77.5\%$$

$$R_c = \frac{U_{c1m}}{I_{c1m}} = \frac{21.6}{0.959} \approx 22.5\Omega$$

5. 【简答题】

Refer to the example at page.69 of the text book, design a Class C power amplifier delivering power of 1W at $f = 500$ MHz with bandwidth is 25 MHz. The conduction angle is $\theta = 45^\circ$ and the power supply voltage is $E_c = 5$ V. The saturation voltage of transistor is $U_{ces} = 0.3$ V.

答案:

The maximum amplitude of the output voltage is

$$U_{c1m} = E_C - U_{ces} = 5 - 0.3 = 4.7V$$

The equivalent collector load resistance is

$$R_c = \frac{U_{c1m}^2}{2P_{out}} = \frac{11^2}{2 \times 5} = 11.045\Omega$$

The amplitude of the output current is

$$I_{c1m} = \frac{U_{c1m}}{R_c} = \frac{20}{47} = 0.426 \text{ (A)}$$

The DC current is

$$I_{c0} = \frac{I_{c1m}}{\gamma_1} = 0.236 \text{ (A)}$$

For $\theta = 45^\circ$, the maximum collector current is

$$I_{cmax} = \frac{I_{c0}}{\alpha_0(45^\circ)} = \frac{0.226}{0.1649} = 1.37 \text{ (A)}$$

The maximum collector-to-emitter voltage is

$$U_{cem} = 2E_C = 2 \times 5 = 10 \text{ (V)}$$

DC power of the supply is

$$P_{DC} = E_C I_{c0} = 5 \times 0.226 = 1.15 \text{ (W)}$$

Power dissipation at the collector is

$$P_C = P_{DC} - P_{out} = 1.15 - 1 = 0.15 \text{ (W)}$$

The collector efficiency is

$$\eta_c = \frac{P_{out}}{P_{DC}} = \frac{1}{1.15} = 87\%$$

The quality factor with load is

$$Q_L = \frac{f}{BW} = \frac{500}{25} = 20$$

The reactance of the resonant circuit component L is

$$X_L = X_C = \frac{R}{Q_L} = \frac{11.045}{20} = 0.55 \text{ (}\Omega\text{)}$$

Yielding

$$L = \frac{X_L}{\omega} = \frac{0.55}{2\pi \times 500 \times 10^6} = 0.176 \text{ (nH)}$$

and

$$C = \frac{1}{\omega X_C} = \frac{1}{2\pi \times 500 \times 10^6 \times 0.55} = 0.576 \text{ (nF)}$$